HW7 - Computer Simulation II

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1 Cluster estimator for the correlation function

For the 2D Ising model we compared the standard estimator

$$G(R) = \langle s_i s_j \rangle$$

with the cluster estimator

$$\hat{G}(i, j) = \frac{V}{|C|} \Theta_C(i) \Theta_C(j),$$

using the Wolff single-cluster algorithm for lattices of size $L = 64$ and $L = 128$.

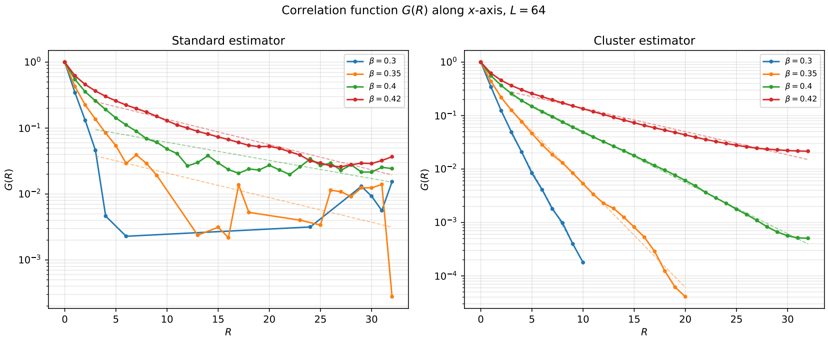

Figure 1: Correlation function $G(R)$ along the x-axis for $L = 64$.

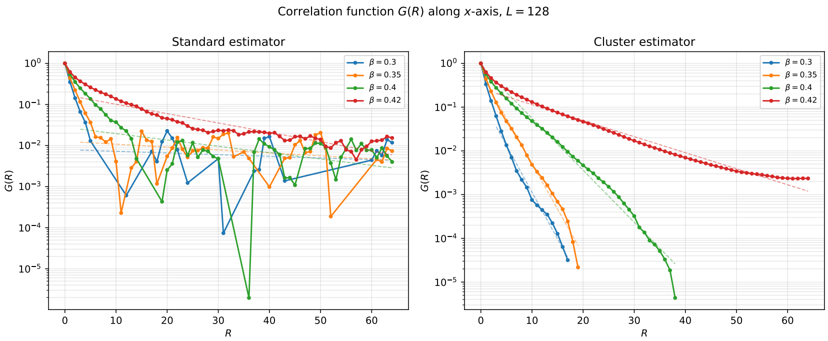

Figure 2: Correlation function $G(R)$ along the x-axis for $L = 128$.

The correlation function decreases exponentially with distance for all temperatures considered,

$$G(R) \propto e^{-R/\xi}.$$

The cluster estimator shows considerably smaller fluctuations than the standard estimator, particularly at large distances where $G(R)$ is small. This leads to a more accurate determination of the correlation length.

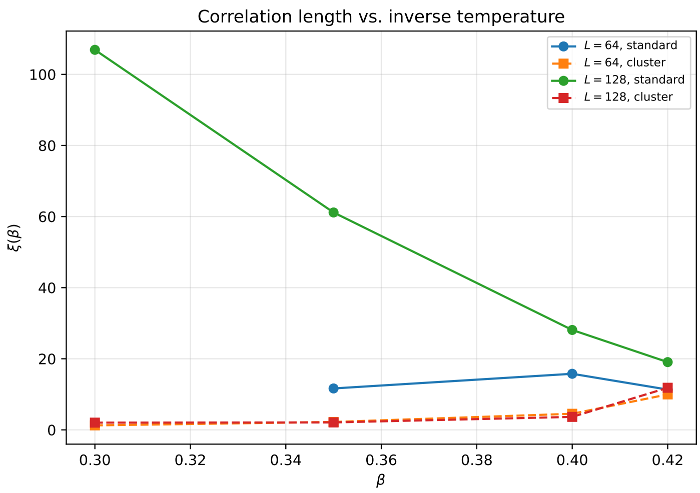

Figure 3: Correlation length as a function of inverse temperature.

As expected, the correlation length increases strongly with β and becomes largest close to the critical point. The results obtained for $L = 64$ and $L = 128$ are consistent and indicate only small finite-size effects in the investigated temperature range.

2 Statistical error of histograms

For a 16×16 lattice at $\beta_c \approx 0.440686$, the Metropolis algorithm was used with 5000 sweeps for thermalisation and $2^{16} = 65536$ sweeps for measurements.

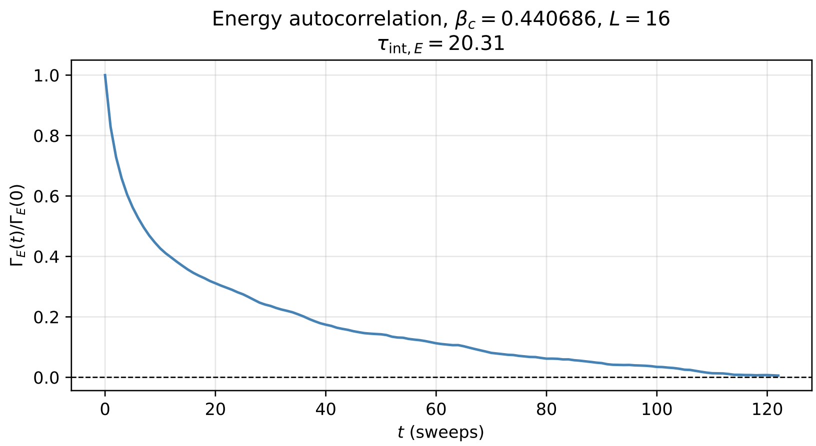

Figure 4: Energy autocorrelation function.

The integrated autocorrelation time obtained from the energy autocorrelation function is

$$\tau_{\mathrm{int},E} = 20.31.$$

The measured energy histogram agrees with the expected distribution at criticality. The histogram errors were obtained from 16 blocks and compared with the Poisson estimate

$$\sigma_{H(E)} \simeq \sqrt{H(E)}.$$

The measured histogram errors are significantly larger than the Poisson prediction due to autocorrelations in the Markov chain. From the measured autocorrelation time we expect an enhancement factor

$$\sqrt{2\tau_{\mathrm{int},E}} = \sqrt{2 \cdot 20.31} = 6.37,$$

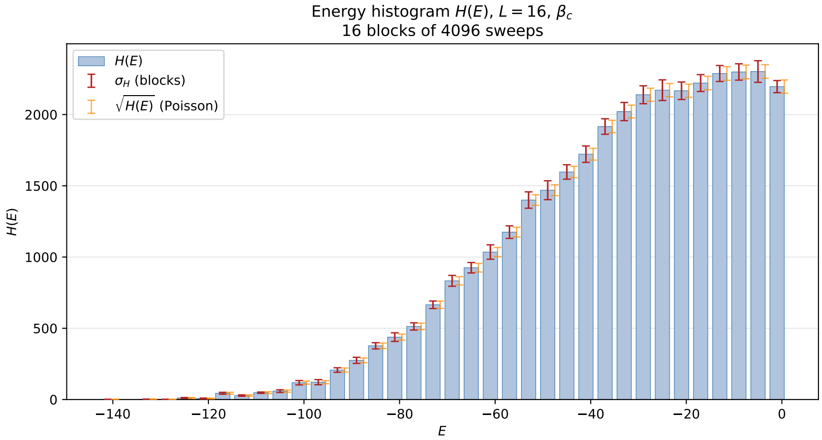

Figure 5: Energy histogram $H(E)$.

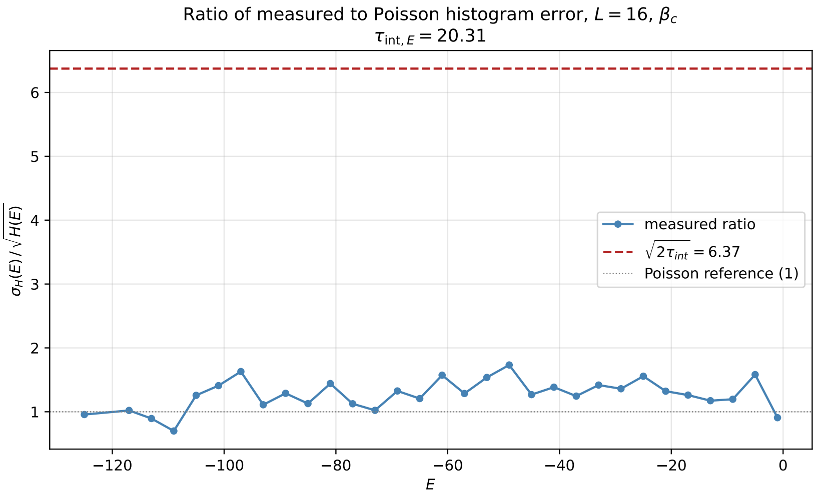

Figure 6: Ratio of measured histogram error to the Poisson estimate.

which is consistent with the observed ratio. Therefore the histogram errors are larger than $\sqrt{H(E)}$ by approximately the expected factor arising from autocorrelated measurements.